Dr. KM Divya Chaturvedi

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Webpage: https://sites.google.com/view/divyachaturvedi
Google Scholar: https://scholar.google.com/citations?user=6zj2ccUAAAAJ&hl=en&authuser=1

Research Interest

- Substrate Integrated Waveguide (SIW) based Cavity-backed Multi-band Antennas with Intrinsically Isolated Ports
- Breast Cancer Detection in Medical Internet of Things
- Smart Antenna Design with IoT-enabled devices
- Leaky Wave antennas
- Phased Array Antenna for 5G Technology
- Fluid antenna for 6G Technology
- Machine Learning Algorithms on medical images

Experience

- Assistant Professor (Regular) at IIIT Pune (July 2024-till date)
- Assistant Professor (Regular) at SRM University-AP, Amravati (Jan. 2020-2024)
- Assistant Professor (Temporary) at IIIT Tiruchirappalli (Jan 2019-Dec. 2019)

Education

Ph.D. (Dept. of Electronics and Communication Engineering, Aug 2015-2019, CGPA: 9)
- National Institute of Technology Trichy (NITT), Tamil Nādu, India.
- Specialization: RF & Microwave

M. Tech. (Dept. of Electronic Engineering Aug. 2013-2015, CGPA: 8.89)
- Pondicherry Central University, India.

B. Tech. (Dept. of Electronics and Communication Engineering, 2007-2011, percentage: 73.44%)
- Institute of Engineering and Technology, affiliated to Uttar Pradesh Technical University, India

Funded Project (ongoing)

- Sponsored by **CSIR** as **Principal Iinvestigator** on "**Design and Development of Substrate Integrated Waveguide (SIW) based Millimeter Wave Beam-Steerable Antenna Array Systems for 5Getworks**", Duration 3 years, outlay **30 Lakhs.**
- Sponsored by **DOT-TTDF** as **Principal Iiinvestigator on "Fluid Dynamics in Wireless: Development of Fluid Antennas for 6G Networks"** Duration 3 years, outlay of 52 Lakhs.
- SERB Power grant project (ongoing) on "**Development of Breast Cancer Detecting System Based on Microwave - Antenna - Array - Sensors and Implementation of Internet of Medical Things (IoMT)**" as **Principal Investigator** of total outlay of **27 Lakhs**
- Completed **Seed-Grant project sponsored by SRM-AP** of total outlay **17 Lakhs** to establish **Anechoic chamber as a Principal investigator.**
- Received an **ISRO response** project on "**Development of Software package for conformal array analysis**" as **Co-PI** with total outlay of **20 Lakhs**

Participation

- Attended more than 20 Conferences / workshops / GIAN Courses/ Seminars / FDP

Special Achievements/

- Received **Best Researcher award** from SRM-AP of total award price **50K in** April 2024
- Cited in **world's top 2% Scientists** declared by Stanford University, **2023** and **2024**, and **2025**.
- Conducted **5-days FDP on "*Recent advancement in RF and Antenna Technologies*"** at IIIT Pune during 16-20th may 2025 as **organizer.**
- Conducted **6-days FDP** on "***Recent Trends in Microwave and Beyond Techniques***" during 15-20th Nov. 2021, at SRM-AP as **organizer.**

Activities/Awards	• Serving as **Senior Member of IEEE** since 2024. • Serving as a **Life senior member** for **WAMS** society since 2024. • Recipient of **PhD Fellowship** from MHRD, Government of India. • Recipient of **Postgraduate Scholarship** (2013-15) from MHRD, Government of India. • Qualified ***'Graduate Aptitude Test in Engineering'*** (GATE – 2013 & 2015) • Received **Young Research award** from ***DST/ SERB, Govt. of India*** to present a research paper in PIERS 2023 at Prague, Czech Republic. • Awardee of Travel **Grant fund (100,000 JPY (*))** from ***Asia Pacific Microwave Conference*** (APMC 2018), Kyoto, Japan. • Received Young Research award from ***DST/ SERB, Govt. of India*** to present research paper in EuCAP 2019, Krakow Poland. • Selected as a winner of the **APMC 2018 Student Travel Grant of 100,000 JPY.** • Received Travel Support from **CSIR** to present a research paper at a conference APMC 2018, Kyoto, Japan. • Received award in CST modelling Challenge in **IAW 2015** at TCE College, Madurai. • Secured rank in top 5% online **NPTEL** certification course on 'ANTENNAS'. • Received the Travel grant from **IMARC 2018**, Kolkata to participate in PhD initiative program. Research Grant: Recipient of EuCAP-2019 Grant (500 Euro) as a Young Researcher. • Best Paper Award **APSYM 2018**: Received Best Paper Award (Rs. 5,000) at APSYM2018 (Dec. 3-5), CUSAT, Cochin. • Travel Grant: Received **Travel Grant** of Rs 5000 from Organizers of IEEE **iAIM 2017, Bangalore** • PhD Students are awarded with PhD initiative awards in the international conference MAPCON 2022, MAPCON 2023, MAPCON 2024.
Journal Publications	Research Credentials Google Scholar Citation: **1252**, h-index: **22**, i10-index: **30** Refereed IEEE & SCI Journals **(33)** As first authors/corresponding author/supervisor**: 26** 1. T. Lanka, **D. Chaturvedi**, MVL Bhavani, A. Kumar "Integration of a Slotted Monopole Antenna with a Partially Reflective Surface for Enhanced Breast Tumor Detection", **IEEE Access**, DOI: 10.1109/ACCESS.2026.3662507 2. **D. Chaturvedi**, MVL Bhavani, A. Kumar "Design of Rectangular Slotted-Patch Antenna Array-Sensor for Breast-Tumor Detection" ***IEEE Latin America Transactions***, vol. 23.no. 11, pp. 1143-1149, 2025. 3. A. Kumar, **D. Chaturvedi**, F. Sarrazin "Characterization and Modelling of Self-Diplexing MIMO Antenna for Endoscopy Capsule Applications" ***International Journal of Communication Systems***, 2025. DOI.org/10.1002/dac.70179 4. T. Lanka, **D. Chaturvedi**, MVL Bhavani, Arvind Kumar "Design of PRS Enabled Monopole Slotted-Antenna Sensor for Breast Tumor Detection" ***Scientific report Journal***, April 2025. DOI: 10.1038/s41598-025-99102-9 5. **D. Chaturvedi**, T. Lanka, A. Kumar "Compact 4-Port MIMO Antenna-Diplexer Utilizing Slotted-HMSIW Technology" ***IEEE Latin America Transaction***, vol. 23, no. 7, pp. 628-635, 2025. 6. D. Chaturvedi, B. Pramodini, T. Lanka "Compact MIMO Antenna with Extended Bandwidth Enabled by Parasitic Patch Structure" ***Journal of Electromagnetic wave and propagation***, April 2025. (Accepted) 7. B. Pramodini, **D. Chaturvedi**, L. Darasi, G. Rana and A. Kumar, "Optimized Compact MIMO Antenna Design: HMSIW-Based and Cavity Backed for Enhanced Bandwidth," IEEE ***Access***, vol. 12, pp. 189820-189828, 2024. 8. MVL Bhavani, **Chaturvedi D**., Lanka G., Kumar A. "Development of a QMSIW Antenna Sensor for Tumor Detection Utilizing a Hemispherical Multilayered Dielectric Breast-Shaped Phantom", ***IEEE Sensor Journal***, Aug. 2024, DOI: 10.1109/JSEN.2024.3450990 9. **Chaturvedi D**., Jadhav P., A, Ayman A "A Compact 2-Port QMSIW Cavity-Backed MIMO Antenna with Varied Frequencies using CSRR-Slot Angles for WBAN Application" ***IEEE Access***, July 2024, https://ieeexplore.ieee.org/document/10589642

10. **Chaturvedi D.,** Kumar A. "A QMSIW Cavity-Backed Self-Diplexing Antenna with Tunable Resonant Frequency Using CSRR Slot" ***IEEE Antennas and Wireless Propagation Letters***. DOI10.1109/LAWP.2023.3323008
11. Pramodini B., **Chaturvedi D.,** Rana G. "Design and Investigation of Dual-Band 2× 2 Elements MIMO Antenna-Diplexer Based on Half-Mode SIW" ***IEEE Access*** 2022, 10, 79272-79280, IF: 3.37
12. **Chaturvedi, D.,** Kumar, A. and Raghavan, S. "A Nested SIW Cavity-Backing Antenna for Wi-Fi/ISM Band Applications" ***IEEE Transactions on Antennas and Propagation***. vol. 67, no. 4, pp. 6, Jan 2019. IF: 4.38
13. **Chaturvedi, D.,** Kumar, A and Raghavan, S. "An Integrated SIW Cavity-Backed Slot Antenna-Triplexer", ***IEEE Antennas and Wireless Propagation Letters***, vol. 17, no. 8, pp 1557-60, 2018. IF: 3.83
14. **Chaturvedi, D.,** Kumar, A. and Raghavan, S. "A Wideband HMSIW based Slotted Antenna for Wi-Fi Application", ***IET Microwave Antenna & Propagation*** 2018 Doi: 10.1049/iet- map.2018.5110 IF: 2.02
15. **Chaturvedi D.,** Kumar A, AA Althuwayb, "Dual-Band Dual-Polarized SIW Cavity-Backed Antenna-Duplexer for Off-body communication" ***Alexandria Engineering Journal*** 2022, IF: 6.77
16. **Chaturvedi D.,** Ayman A "Bandwidth enhancement of a planar SIW cavity-backed slot antenna using slot and metallic shorting via" Applied Physics A, 2022. IF: 2.98
17. Kumar A., **Chaturvedi D.,** Rosaline I "Design of Antenna-Multiplexer for Seamless On-Body Internet of Medical Things (IoMT) Connectivity" ***IEEE Transactions on Circuits and Systems II: Express Briefs*** 2022. DOI 10.1109/TCSII.2022.3170513, IF: 3.69
18. Ayman A, **Chaturvedi D.** "Substrate Integrated Waveguide Based Cavity-Backed Bowtie-Slot Antenna-Triplexer", ***International Journal of RF and Microwave Computer-Aided Engineering***, 2021. DOI: 10.1002/mmce.22520, IF: 1.7
19. Ayman A., Al_Hasan, M., **Chaturvedi, D**. "Design of half-mode substrate integrated cavity inspired dual-band Antenna" ***International Journal of RF and Microwave Computer-Aided Engineering***, 2021. DOI: 10.1002/mmce.22520, IF: 1.7
20. **Chaturvedi, D.,** "SIW Cavity-Backed 24o Inclined-Slots Antenna for ISM Band Application" ***International Journal of RF and Microwave Computer-Aided Engineering,*** 2020.
21. **Chaturvedi, D.** "Substrate Integrated Waveguide Based Cavity-Backed Bowtie-Slot Antenna-Triplexer" ***International Journal of RF and Microwave Computer-Aided Engineering*** 2021.
22. **Chaturvedi, D.** and Raghavan, S. "A dual-band half-mode substrate integrated waveguide- based antenna for WLAN/WBAN applications", ***International Journal of RF and Microwave Computer-Aided Engineering*** 2019, vol. 28, no. 5, pp 1-9, 2017.
23. **Chaturvedi, D.** and Raghavan, S., "Compact QMSIW-based antenna with different resonant frequencies depending on loading of metalized vias" ***International Journal of Microwave and Wireless Technologies*** 2019, Vol. 11, no. 4, pp. 420-427, 2019.
24. **Chaturvedi, D.,** Raghavan S. "A Compact Metamaterial-Inspired Antenna for WBAN Application", ***Wireless Personal Communications*** 2018, Vol. 105, no. 4, pp 1449–1460, 2019. DOI: 10.1007/s11277-019-06153-z. IF: 2.64
25. **Chaturvedi, D.** and Raghavan, S., Circular Quarter-Mode SIW Antenna for WBAN Application. ***IETE Journal of Research***, vol.64, no. 4, pp. 482-8. IF: 2.33, 2018.
26. **Chaturvedi, D.,** Raghavan S., 2017. A Half-Mode SIW Cavity-Backed Semi-Hexagonal Slot Antenna for WBANApplication. ***IETE Journal of Research***. Doi.org/10.1080/03772063.2018.1452644. IF: 2.33
27. **Chaturvedi, D.,** Raghavan S. "A triangular-shaped quarter-mode substrate integrated waveguide- based antenna for WBAN applications", ***Defence Science Journal***, vol. 68, no.2, pp. 190-195, 2018.
28. Kumar, A., **Chaturvedi, D.,** and Raghavan, S. "Design and Experimental Verification of Dual-Fed, Cavity-Backed Antenna-Diplexer using HMSIW Technique", ***IET Microwaves Antennas & Propagation***, vol. 13, no. 3, pp. 380 – 385, 2019. IF: 2.02
29. Kumar, A., **Chaturvedi, D.,** and Raghavan, Singaravelu "Design of a self-diplexing antenna using SIW technique with high isolation", ***Elsevier AEU-International Journal of Electronics and Communications***, 94, 386-391, 2018. IF: 3.18
30. Kumar, A., **Chaturvedi, D.,** and Raghavan, SIW cavity-backed circularly polarized square ringslot antenna with wide axial-ratio bandwidth", ***Elsevier AEU-International Journal of Electronics and Communications***, 94, 386-391, 2018. IF: 3.18
31. Ayman A, **Chaturvedi D.** et. al. "Substrate integrated waveguide (SIW) cavity-backed slot antenna with monopole-like radiation for vehicular communications" ***Applied Physics A***, 2022. DOI: 10.1007/s00339-021-05203-3, IF: 2.98
32. **Chaturvedi, D.,** Raghavan S. "A triangular-shaped quarter-mode substrate integrated waveguide-based antenna for WBAN applications", ***Defense Science Journal***, vol. 68, no.2, pp. 190-195, 2018.

	33. **Chaturvedi, D.**, Raghavan S. "A Compact Metamaterial-Inspired Antenna for WBAN Application", ***Wireless Personal Communications*** 2018, Vol. 105, no. 4, pp 1449–1460, 2019. DOI: 10.1007/s11277-019-06153-z
Conference Publications	**Total conference papers: National / international** published in IEEE Xplore: **(24),** Foreign conferences **(14)** 1) **D. Chaturvedi**, T. Ganesh, A. Kumar "Compact Design of a 4-Port MIMO Antenna-Diplexer Leveraging HMSIW Cavities" IEEE European Conference on Antenna and Propagation (EuCAP), Stockholm, Sweden, March 30-April 4th 2025. (***Accepted***) 2) A. Kumar, **D. Chaturvedi**, et.al. "QMSIW Based Self-Diplexing MIMO Antenna" IEEE European Conference on Antenna and Propagation (EuCAP), Stockholm, Sweden, March 30-April 4th 2025. (***Accepted***) 3) T. Ganesh, **D. Chaturvedi**, M. Bhavani, A. Kumar "Design of Meta Surface Based Monopole Antenna Sensor for Breast Cancer Detection" 2024, IEEE Microwaves, Antennas, and Propagation Conference (MAPCON), Hyderabad, Dec 9-13. 4) M. Bhavani, **D. Chaturvedi**, A. Kumar "A QMSIW Based Antenna Sensor for Breast Tumor Detection" IEEE Microwaves, Antennas, and Propagation Conference (MAPCON), Hyderabad, Dec 9-13, 2024. 5) T Ganesh, **D Chaturvedi**, "Design of Wearable Monopole Antenna Sensor for Breast Tumor Detection" 2024 IEEE Wireless Antenna and Microwave Symposium (WAMS), 1-4, Vishakhapatnam. 6) B Pramodini, **D Chaturvedi**, "A Half-Mode SIW Cavity-Backed 3-Ports MIMO Antenna", 2024 IEEE Wireless Antenna and Microwave Symposium (WAMS), 1-4, Vishakhapatnam. 7) S Sharma, **D Chaturvedi** "Slotted Rectangular Microstrip Patch Antenna for Breast Cancer Detection", 2023 Photonics & Electromagnetics Research Symposium (PIERS), 557-561, Prague, Czech Republic. 8) **D Chaturvedi**, T Ganesh, A Kumar "Design and Analysis of Wearable Monopole Antenna Sensor," 2023 Photonics & Electromagnetics Research Symposium (PIERS), 2003-2007, Prague, Czech Republic. 9) S Imamvali, R Chaparla, S Tupakula, **D Chaturvedi** "Novel SSPP Sensor System with Octagon-shaped Unit Cell for Liquid Analyte Dielectric Constant Detection," 2023 Photonics & Electromagnetics Research Symposium (PIERS), 1467-147, Prague, Czech Republic. 10) R Kumar, A Chandra**, D Chaturvedi** "Wideband circularly polarized rectangular dielectric resonator antenna using inverted U-shaped ground plane for sub 6GHz and upper mid-bands 5G applications", 2023 Photonics & Electromagnetics Research Symposium (PIERS), Prague, Czech Republic. 11) B Pramodini, **D Chaturvedi** "Miniaturized SIW-based Cavity-Backed Antenna" 2023 IEEE Wireless Antenna and Microwave Symposium (WAMS), 1-4. 12) PM Neelamraju, P Pothapragada, G Rana, **D Chaturvedi**, R Kumar "Machine Learning based Low-Scale Dipole Antenna Optimization using Bootstrap Aggregation", 2023 2nd International Conference on Paradigm Shifts in Communication, VNIT Nagpur. 13) B Pramodini, **D Chaturvedi** "Miniaturized Inverted L-Shaped Slot Antenna Using HMSIW Technology", 2023 2nd International Conference on Paradigm Shifts in Communications, VNIT Nagpur. 14) **D Chaturvedi**, B Pramodini, G Rana, A Kumar, "Design of a Compact HMSIW Cavity-Backed Dual-Band 4-Port MIMO Antenna", 2022, 27th Asia Pacific Conference on Communications (APCC), 145-149, Jeju, South Korea. 15) A Kumar, **D Chaturvedi**, DS Rajput, S Sivanantham "Antenna-multiplexer for IoMT wireless connectivity", 2022 27th Asia Pacific Conference on Communications (APCC), 371-374, Jeju, South Korea. 16) **D Chaturvedi** "SIW Cavity-Backing Antenna-Diplexer using Nested Cavities", 2021 IEEE Indian Conference on Antennas and Propagation (InCAP), 486-489, 2021, MNIT Jaipur. 17) A Kumar, **D Chaturvedi**, AA Khan "Bandwidth-Improved HMSIW Resonator Based Filtennas for Single/Dual Channel Network",2021 IEEE Indian Conference on Antennas and Propagation (InCAP), 502-505, MNIT Jaipur. 18) M Kishore Babu, **D Chaturvedi**, G Rana "Substrate Integrated Waveguide Based Periodic Leaky Wave Antenna with Low Stopband", 2021 IEEE Indian Conference on Antennas and Propagation (InCAP), 1027-1030, MNIT Jaipur. 19) **D Chaturvedi**, A Kumar, S Raghavan" A planar SIW cavity-backed tilted-slots antenna for WBAN application", 13th European Conference on Antennas and Propagation (EuCAP), 1-5, 4, 2019, Krakow, Poland.

	20) A Kumar, **D Chaturvedi**, S Raghavan "Dual-band, dual-fed self-diplexing antenna", 2019 13th European conference on antennas and propagation (EuCAP), 1-5, Krakow, Poland. 21) A Kumar, **D Chaturvedi**, M Saravana Kumar, S Raghavan "SIW cavity-backed self-triplexing antenna with T-shaped slot", 2018 Asia-Pacific Microwave Conference (APMC), 1588-1590, 2018, Kyoto, Japan. 22) **D Chaturvedi**, A Kumar, S Raghavan "A substrate integrated waveguide-based antenna-triplexer", 2018 Asia-Pacific Microwave Conference (APMC), 10-12, Kyoto, Japan. 23) **D Chaturvedi**, S Raghavan "A quarter-mode SIW based antenna for ISM band application", 2017 IEEE International Conference on Antenna Innovations & Modern, 2017, ISRO, Bangalore. 24) **D Chaturvedi**, S Raghavan "On-body resilient SIW based antenna for WBAN applications", NCC, IIT Madras.
Patent	• **Divya Chaturvedi**, Tiruganesh "Design and Analysis of Miniaturized Patch Antenna-Array for Ground Penetrating Radar Application" (published), Application No. 202441007753 • **Divya Chaturvedi**, Arvind Kumar, Ashish Chandelkar, "An Octa-Band Self-Multiplexing Antenna (SMA) For Sub 6ghz Applications", Application No. 202441039762 (published), SRM University-AP
Professional Activities	**Reviewer:** • IET-MAP, AWPL, Elsevier (AEU), ACES Journal, Willey RF-CAD, EUMA CambridgePress (EUMA), MOTL. • APMC 2019, APMC 2022 **Membership:** IEEE –APS Member (Jan. 2016 - till date), IEEE Microwave Theory and Techniques, MTT, IEEE Women in Engineering
Ph.D. guide	**Ph.D. Supervisor** • **Avinash Pandey "**Design of SIW-millimeter Antennas based system for 5G Network" July 2025 till date. • **MVL Bhavani** "Breast Cancer detection using Microwave Imaging" Supervisor, Oct 2023 till date. • **Tiruganesh Lanka** "Design of Wearable Antenna-Array Sensor for WBAN" S***upervisor,*** June. 2022 till date. • **Buela Pramodini** "Design of SIW based Cavity-backed MIMO Antennas" ***Supervisor,*** May 2021 till date (thesis submitted)
Technical Expertise	• **Electromagnetic Simulation Software:** CST Microwave Studio, Ansoft HFSS, Zealand IE3D, MATLAB, AWR for analog circuit & system design • **PCB fabrication:** Planar antennas, Substrate Integrated Waveguide antenna and Microwave circuits. (Expertise in using MITS Prototyping machines and screen-printing technology) • **Equipment & Instruments:** Vector Network Analyzer, Spectrum Analyzer, TRL calibration kit, Antenna Turn table in radiation pattern Measurements, Microwave test Bench. Material Characterization using 2D Lines Methods & NRW, Anechoic chamber, NI Elvis kit
Professional Activities	• 2024: TPC member for MAPCON 2024, Hyderabad • 2024: TPC member for APMC 2024, Bali, Indonesia • 2022: TPC member of IEEE APMC 2022, Yokohama, Japan. • 2021: TPC member of Asia Pacific Microwave Conference 2021 Brisbane, Australia, 2019 Singapore
Subjects Taught	• Transmission lines and Waveguide, Antenna and Propagation , Electromagnetic theory, Microwave Theory and Engineering, RF Microwave Imaging, Basic Electrical and Electronics, Digital Electronics •
Academic Responsibilities	• ECE Dept. academic coordinator • Department Data updating coordinator, admission committee member Jan. 2020-Dec. 2020 Divya